

CS-23334-FUNDAMENTALS OF DATA SCIENCE

AMAR GANESH S 240701035

Experiment No: 2

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2.Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Visualize the following:

1. Sales over the product
2. Sales over time
3. Display the correlation matrix

AIM:

To analyze the given data and perform preprocessing and visualize the data as Bar Plot , Line Plot , Pivot Table and a Correlation Matrix

Algorithm:

Step 1: Data Loading and Preprocessing

Step 2: Data Visualization (Bar Plot, Line Plot, Pivot Table, Correlation Matrix)

Step 3: Interpretation and Reporting

About Dataset:

The dataset contains daily sales records for three products across different regions, including information on sales amount, quantity sold, and transaction date.

Code With Output:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# STEP 1: Upload the file manually
from google.colab import files
uploaded = files.upload()

# STEP 2: Load the uploaded file into a pandas DataFrame
# (Make sure the file name matches exactly what you upload, e.g.,
# 'Sales_Data.csv')
df = pd.read_csv('Sales_Data.csv')

# STEP 3: Display the first few rows
print(df.head())

# STEP 4: Check for missing values
print("\nMissing Values:\n", df.isnull().sum())

# STEP 5: Handle missing data
df['Sales'] = pd.to_numeric(df['Sales'], errors='coerce')
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# STEP 6: Summary statistics
print("\nSummary Statistics:\n", df.describe())

# STEP 7: Group by product and calculate total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print("\nProduct Summary:\n", product_summary)

# STEP 8: Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# STEP 9: Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
sales_over_time = df.groupby('Date').agg({'Sales':
    'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.tight_layout()
plt.show()
```

```

# STEP 10: Pivot table - Sales by Region and Product
pivot_table = df.pivot_table(values='Sales', index='Region',
columns='Product', aggfunc=np.sum, fill_value=0)
print("\nPivot Table:\n", pivot_table)

# STEP 11: Correlation matrix
correlation_matrix = df.select_dtypes(include=[np.number]).corr()
print("\nCorrelation Matrix:\n", correlation_matrix)

# STEP 12: Heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()

```

Output:

<IPython.core.display.HTML object>

Saving Sales_Data.csv to Sales_Data (1).csv

	Date	Product	Sales	Quantity	Region
0	01-01-2023	Product A	200	4	North
1	02-01-2023	Product B	150	3	South
2	03-01-2023	Product A	220	5	North
3	04-01-2023	Product C	300	6	East
4	05-01-2023	Product B	180	4	West

Missing Values:

Date	0
Product	0
Sales	0
Quantity	0
Region	0

dtype: int64

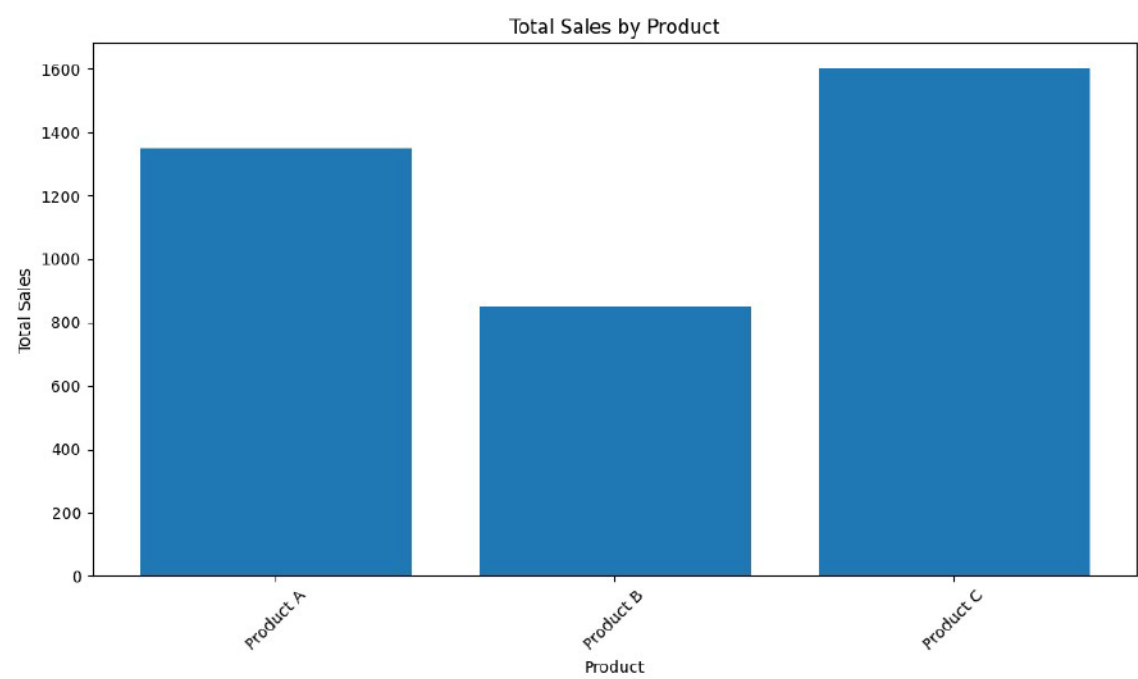
Summary Statistics:

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

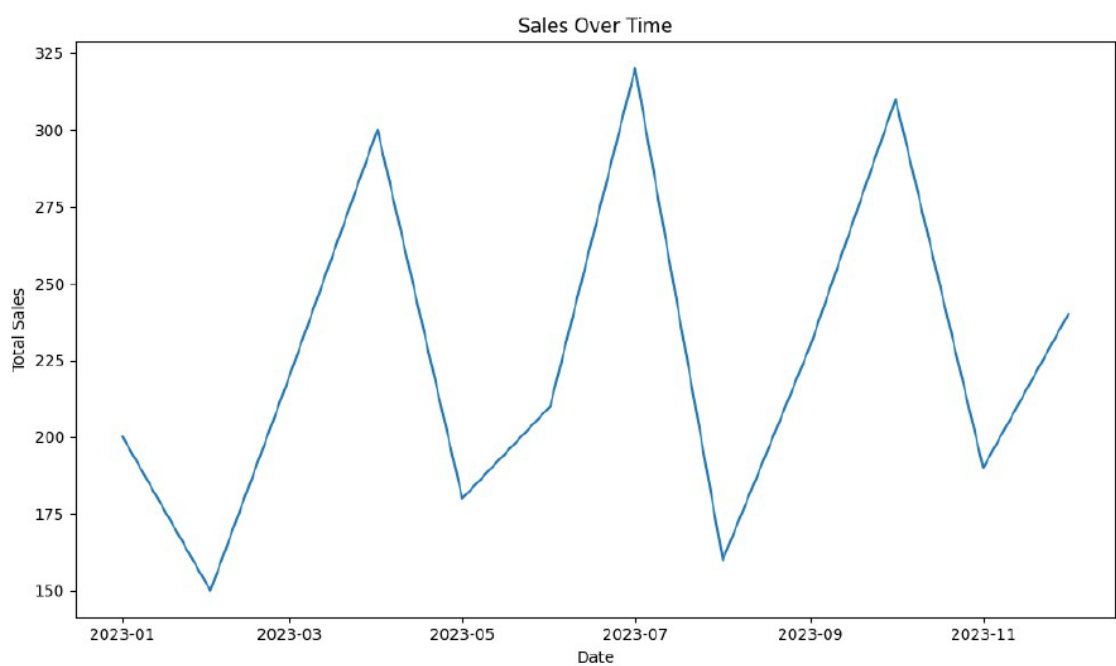
Product Summary:

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36

Bar Plot:



Line Plot:



Pivot Table:

Pivot Table:

Product	Product A	Product B	Product C
Region			

East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

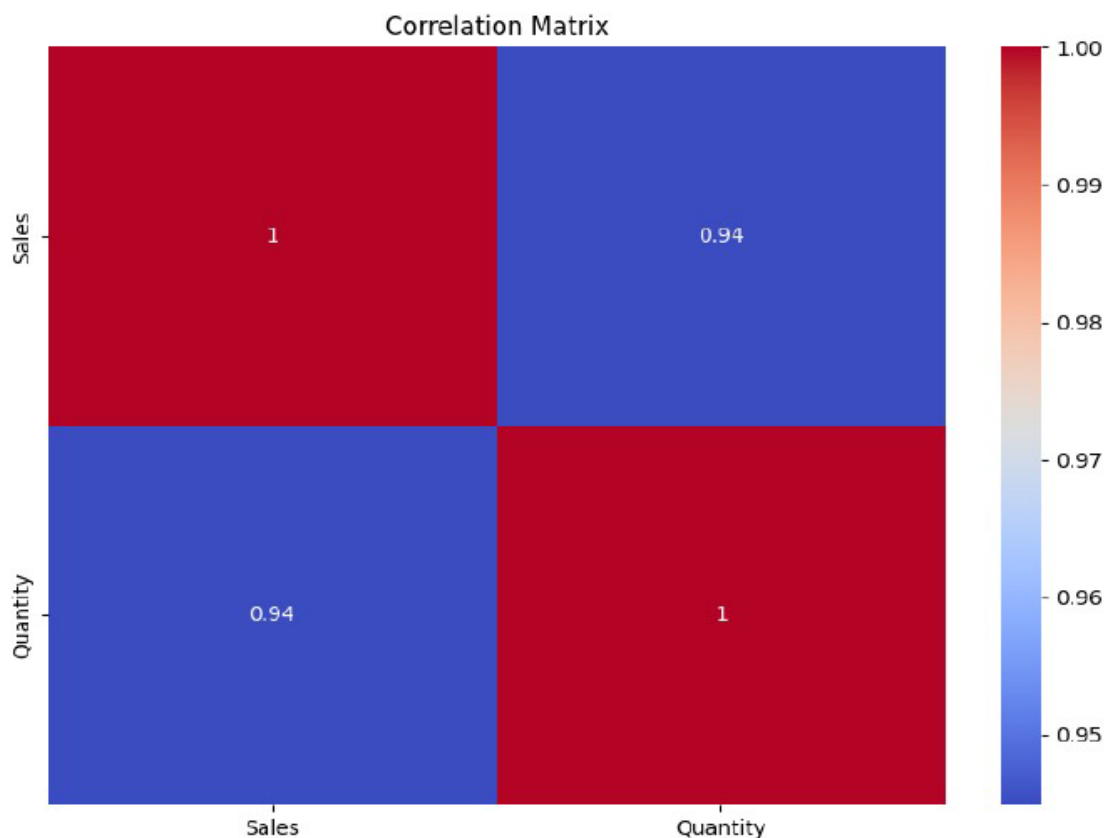
Correlation Matrix:

Correlation Matrix:

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000

/tmp/ipython-input-217718739.py:57: FutureWarning: The provided callable <function sum at 0x7a4182919620> is currently using DataFrameGroupBy.sum. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "sum" instead.

```
pivot table = df.pivot table(values='Sales', index='Region',  
columns='Product', aggfunc=np.sum, fill_value=0)
```



Result:

Thus the given dataset was preprocessed and visualized using a python code.